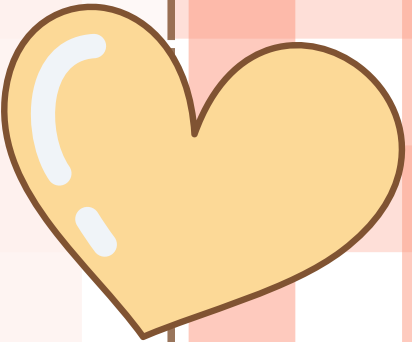


Magnetism

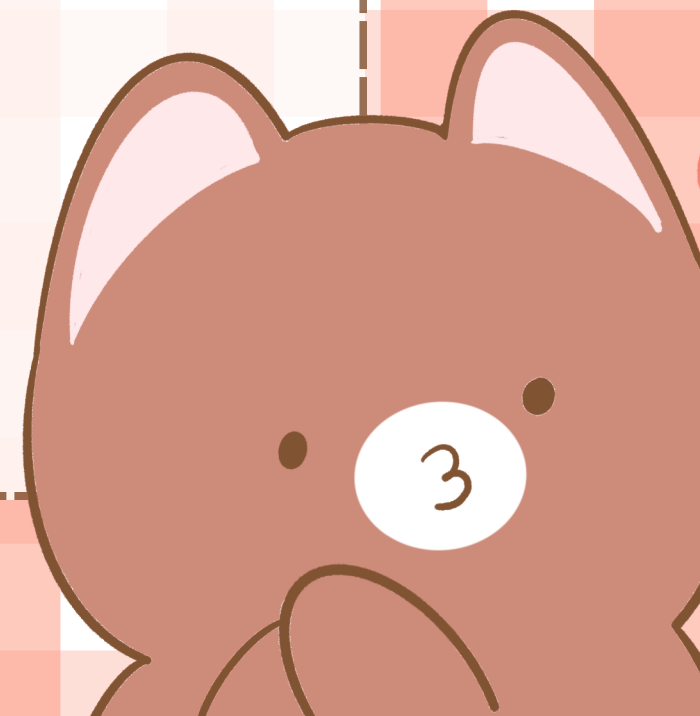
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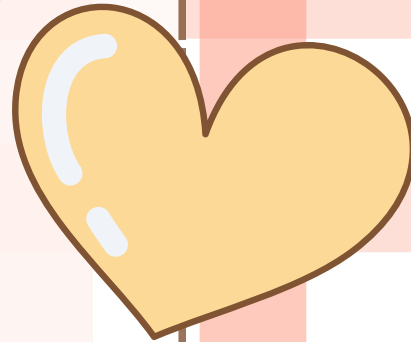




Definition

Magnetism is associated with magnetic fields, which arise from the motion of electric charges. This motion can take many forms.

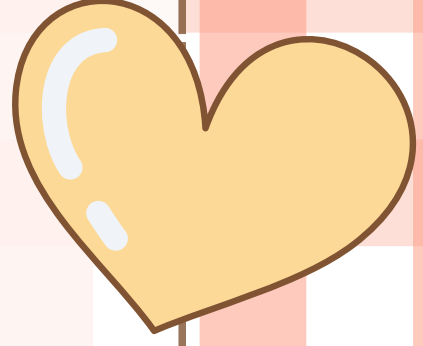
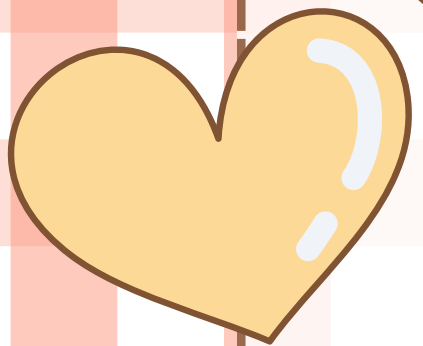




Magnetic Fields

Evidence for the presence of a magnetic field is the magnetic force on charges moving in that field; the force is at right angles to both the field and the velocity of the charge. This force deflects the particles without changing their speed.

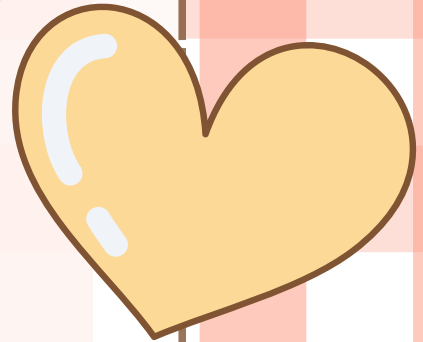




Types of Magnets

- Permanent Magnets: Materials that maintain their magnetic properties for a long time.
- Temporary Magnets: Materials, like a nail, that become magnetized in the presence of a permanent magnet but may lose their magnetism.
- Electromagnets: Created by electric currents flowing through a wire, often coiled around an iron core to increase strength.

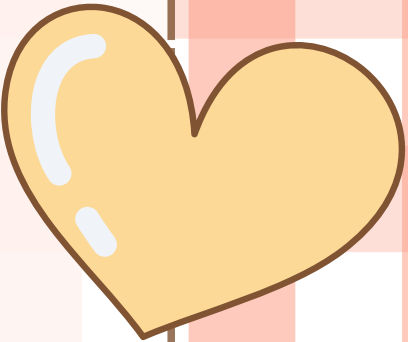




Examples

When a magnet bar is brought close to a paperclip, the paperclip is magnetized to the bar. . The magnetic field of the bar magnet causes the magnetic domains within the steel paperclip to align, turning it into a temporary magnet that is attracted to the bar magnet.





Formulas

- Lorentz Force : $\text{Force} = ((\text{electric charge of particle} / q) \times (\text{electric field vector} / E) + \text{electric charge of particle} / q \times ((\text{velocity vector of particle} / v) \times (\text{magnetic field vector} / B))$
- Right hand rule: if the right-hand fingers follow the curve of a point on the object, then the thumb points along the axis of rotation in the direction of the angular velocity vector.





Thank you

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